

<MICRO PROJECT USING GUI FOR STUDENT ADMISSION FORM>

Submitted by

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1.INTRODUCTION

PROBLEM STATEMENT

Create a GUI for student's admission form. It must contain first name, middle name, last name, date of birth, gender (options will be there), contact and permanent address with pin code, locality, city, district, PO, PS, urban or not, state, country, email, phone no, options for veg, non-veg, name of parents and guardian along with their detailed address occupation, phone number, email, class 12 subjects with marks, % of PCM marks will be calculated automatically, hobbies, etc. The form must contain horizontal and vertical scroll bar. Save the information in a file.

PROBLEM DEFINITION

In this provided microproject on the topic of creation of a Graphical User Interface (GUI) on a student's admission form, we have gathered a collection of the personal details of the respective students. Various types of characteristic attributes such as name Date of Birth, personal details, phone number, percentage marks of PCM in class 12 have been provided and can be automatically calculated. Henceforth, a quite accessible and easy-going methodological procedure has been created, so that any kind of personal data can be gained and retrieved respectively and vice-versa.

OBJECTIVE

The objective of creating a GUI for an admission form microproject is to design a user-friendly interface that allows students to easily input their personal and academic information for the admission process. The GUI should include all the necessary fields mentioned earlier, such as name, date of birth, contact details, address, parent/guardian information, academic records, and hobbies. Additionally, the GUI should incorporate horizontal and vertical scroll bars to accommodate all the form fields and ensure easy navigation. The ultimate goal is to create a visually appealing and functional interface that simplifies the admission form submission process for students.

BACKGROUND STUDY

Before starting the microproject to create a GUI for a student admission form, it is important to conduct a background study to gather relevant information and understand the requirements. Here are some key areas to focus on during the background study:

1. Admission Process: Familiarize yourself with the admission process of the educational institution or organization for which you are creating the form. Understand the information they require from students during the admission process.
2. Form Fields: Identify all the necessary fields that need to be included in the admission form. This includes personal details (name, date of birth, gender), contact information (address, phone number, email), academic records (subjects, marks), parent/guardian details, and any additional information required.
3. Validation: Research the validation rules and requirements for each form field. Determine any specific formats, character limits, or data constraints that need to be enforced to ensure accurate and valid data entry.
4. User Experience: Consider the user experience aspect of the form. Research best practices for form design, such as using clear labels, grouping related fields, providing helpful instructions, and ensuring easy navigation through the form.
5. Scroll Bars: Understand how to implement horizontal and vertical scroll bars in the GUI framework or programming language you are using. Research the documentation or tutorials specific to the GUI framework to learn how to incorporate scroll bars effectively.

6. File Handling: Research the file handling mechanisms available in the chosen programming language or framework. Understand how to write the form data to a file, including the format and structure of the saved information.

BRIEF DISCUSSION ON THE PROBLEM

The problem at hand is to create a GUI for a student's admission form that encompasses various fields and requirements. The form needs to capture essential personal information such as first name, middle name, last name, date of birth, and gender. It should also include contact details like phone number and email, as well as a complete permanent address with pin code, locality, city, district, PO, PS, urban or not, state, and country. Furthermore, the form should provide options for dietary preferences, allowing students to choose between veg and non-veg. It should also gather information about the student's parents and guardian, including their names, detailed addresses, occupations, phone numbers, and email addresses. Academic records play a crucial role in the admission process, so the form should include fields for class 12 subjects and corresponding marks. Additionally, the form should automatically calculate the percentage of PCM (Physics, Chemistry, Mathematics) marks. To ensure a comprehensive profile, the form should also include a section for students to list their hobbies and interests. To accommodate all the required fields, the form should have both horizontal and vertical scroll bars, allowing users to navigate through the form easily. Lastly, the information provided in the form needs to be saved in a file for future reference and processing. By addressing these requirements and designing a user-friendly interface, the GUI for the student's admission form will streamline the admission process and provide an efficient means of collecting and storing student information.

2.CONCEPT AND PROBLEM ANALYSIS:

The concept of creating a GUI for an admission form involves designing a graphical user interface that allows students to input their personal and academic information in a structured manner. The GUI should provide a user-friendly experience, making it easy for students to navigate through the form and enter the required details. It should include various fields such as name, date of birth, contact information, address, parent/guardian details, academic records, and hobbies. The form should also incorporate horizontal and vertical scroll bars to accommodate all the necessary fields and ensure that the form remains accessible even if the content exceeds the visible area.

The problem at hand is to create a GUI for an admission form that addresses the following challenges:

1. User-Friendly Design: The GUI should be intuitive and easy to use, ensuring that students can navigate through the form without confusion or difficulty.
2. Comprehensive Information: The form should include all the necessary fields to collect relevant information from the students, such as personal details, contact information, address, parent/guardian details, academic records, and hobbies.
3. Validation and Error Handling: The form should incorporate validation mechanisms to ensure that the entered data is accurate and complete. It should also handle errors gracefully, providing appropriate feedback to the user in case of any mistakes or missing information.
4. Scrollable Interface: As the form may contain a significant amount of information, the GUI should include horizontal and vertical scroll bars to accommodate all the fields and allow users to access and fill in the entire form.
5. Data Storage: The form should have a mechanism to save the entered information in a file or database for future reference and processing.

3.CODES AND SCREENSHOTS

CODE:

```
from tkinter import *
import tkinter as tk
from tkinter import ttk
from tkinter.ttk import Combobox
from tkinter import messagebox
import openpyxl,xlrd
from openpyxl import Workbook
import pathlib
file=pathlib.Path('MicroProject1_data.xlsx')
if file.exists():
    pass
else:
    file=Workbook()
    sheet=file.active
    sheet['A1']="Name"
    sheet['B1'] = "Date Of Birth"
    sheet['C1']="Gender"
    sheet['D1']="Address"
    sheet['E1']="E-Mail"
    sheet['F1'] = "Phone Number"
    sheet['G1'] = "Deitary Preference"
    sheet['H1'] = "Father's Name"
    sheet['I1'] = "Mother's Name"
    sheet['J1'] = "Gaurdian Name"
    sheet['k1'] = "Parents' Address"
```

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```
sheet['L1'] = "Parents' E-mail"
sheet['M1'] = "Parents' Phone Number"
sheet['N1'] = "Father's Occupation"
sheet['O1'] = "Mother's Occupation"
sheet['P1'] = "English Marks"
sheet['Q1'] = "Physics Marks"
sheet['R1'] = "Chemistry Marks"
sheet['S1'] = "Maths Marks"
sheet['T1'] = "Optional 1 Marks"
sheet['U1'] = "Aggregate Marks(PCM)"
sheet['V1'] = "Hobbies(Optional)"
sheet['W1'] = "Achievement(Optional)"
file.save('MicroProject1_data.xlsx')

def submit():
    f = fnameVal.get()
    m = mnameVal.get()
    l = lnameVal.get()
    completeName=(f+" "+m+" "+l)
    birthDate=dob.get()
    gen=gender.get()
    postinNo=pin.get()
    locality=loc.get()
    district=dist.get()
    stateName=state.get()
    coun=country.get()
    add=(locality+" , "+"P.I.N.="+postinNo+" , "+"district+" , "+"stateName+" , "+"coun)
    email=emAd.get()
    phone=phn.get()
    dietPref=diet.get()
```


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```
p_postinNo =p_pin.get()
p_locality =p_loc.get()
p_district =p_dist.get()
p_stateName=p_state.get()
p_coun = p_country.get()
p_add = (p_locality + " , " + "P.I.N.=" + p_postinNo + " , " + p_district + " , " + p_stateName + " , " +
p_coun)
p_email = p_emAd.get()
p_phone = p_phn.get()
f_occu=foccu.get()
m_occu=moccu.get()
fatherName=father.get()
motherName=mother.get()
gaurdian=gaurd.get()
phy=p.get()
chem=c.get()
maths=mth.get()
aggMarks=((phy+chem+maths)/3)
eng=e.get()
opt=op.get()
myHob=hobbies.get()
myAch=achieve.get()
file=openpyxl.load_workbook('MicroProject1_data.xlsx')
sheet=file.active
sheet.cell(column=1,row=sheet.max_row+1,value=completeName)
sheet.cell(column=2, row=sheet.max_row, value=birthDate)
sheet.cell(column=3, row=sheet.max_row, value=gen)
sheet.cell(column=4, row=sheet.max_row, value=add)
sheet.cell(column=5, row=sheet.max_row, value=email)
sheet.cell(column=6, row=sheet.max_row, value=phone)
```

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```
sheet.cell(column=7, row=sheet.max_row, value=dietPref)
sheet.cell(column=8, row=sheet.max_row, value=fatherName)
sheet.cell(column=9, row=sheet.max_row, value=motherName)
sheet.cell(column=10, row=sheet.max_row, value=gaurdian)
sheet.cell(column=11, row=sheet.max_row, value=p_add)
sheet.cell(column=12, row=sheet.max_row, value=p_email)
sheet.cell(column=13, row=sheet.max_row, value=p_phone)
sheet.cell(column=14, row=sheet.max_row, value=f_occu)
sheet.cell(column=15, row=sheet.max_row, value=m_occu)
sheet.cell(column=16, row=sheet.max_row, value=eng)
sheet.cell(column=17, row=sheet.max_row, value=phy)
sheet.cell(column=18, row=sheet.max_row, value=chem)
sheet.cell(column=19, row=sheet.max_row, value=maths)
sheet.cell(column=20, row=sheet.max_row, value=opt)
sheet.cell(column=21, row=sheet.max_row, value=aggMarks)
sheet.cell(column=22, row=sheet.max_row, value=myHob)
sheet.cell(column=23, row=sheet.max_row, value=myAch)
file.save(r'MicroProject1_data.xlsx')
```

```
def clear():
```

```
    fnameVal.set("")
    mnameVal.set("")
    lnameVal.set("")
    dob.set("")
    pin.set("")
    loc.set("")
    dist.set("")
    state.set("")
    country.set("")
    emAd.set("")
    phn.set("")
```

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```
father.set("")
mother.set("")
gaurd.set("")
p_pin.set("")
p_loc.set("")
p_dist.set("")
p_state.set("")
p_country.set("")
p_emAd.set("")
p_phn.set("")
foccu.set("")
moccu.set("")
p.set("")
c.set("")
mth.set("")
e.set("")
op.set("")
hobbies.set("")
achieve.set("")

def on_configure(event):
    canvas.configure(scrollregion=canvas.bbox('all'))

window=Tk()
window.title("Admission From")
canvas=Canvas(window)
canvas.pack(side=LEFT,fill=BOTH,expand=True)
vs=Scrollbar(window,orient="vertical",command=canvas.yview,bg="lightgray",troughcolor="gray",
activebackground="darkgray")
vs.pack(side=RIGHT,fill=Y)
hs=Scrollbar(window,orient="horizontal",command=canvas.xview,bg="lightgray",troughcolor="gray",
activebackground="darkgray")
```

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```
hs.pack(side=BOTTOM,fill=X)
canvas.configure(yscrollcommand=vs.set,xscrollcommand=hs.set)
frame=Frame(canvas)
canvas.create_window((0,0),window=frame,anchor="nw")
frame.bind("<Configure>", on_configure)
fnameVal=StringVar()
mnameVal=StringVar()
lnameVal=StringVar()
dob=StringVar()
pin=StringVar()
loc=StringVar()
dist=StringVar()
state =StringVar()
country=StringVar()
emAd =StringVar()
phn =StringVar()
father =StringVar()
mother =StringVar()
gaurd =StringVar()
p_pin =StringVar()
p_loc =StringVar()
p_dist=StringVar()
p_state =StringVar()
p_country=StringVar()
p_emAd =StringVar()
p_phn =StringVar()
foccu =StringVar()
moccu =StringVar()
p =IntVar()
c =IntVar()
```

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```

mth=IntVar()
e =IntVar()
op =IntVar()
hobbies=StringVar()
achieve=StringVar()
Label(frame,text='FillouttheAdmissionForm',font="arial",bg="#eba834",fg="#eb3434").grid(row=0,column
=0)
Label(frame,text="Student's Details",fg="blue").grid(row=5,column=0,pady=30)
fname=Label(frame,text='First Name')
fname.grid(row=6,column=0)
e1=Entry(frame,textvariable=fnameVal,width=50)
e1.grid(row=6,column=1,pady=10)
mname=Label(frame,text='Middle Name')
mname.grid(row=7,column=0)
e1=Entry(frame,textvariable=mnameVal,width=50)
e1.grid(row=7,column=1,pady=5)
lname=Label(frame,text='Last Name')
lname.grid(row=8,column=0)
e1=Entry(frame,textvariable=lnameVal,width=50)
e1.grid(row=8,column=1,pady=5)
dobLabel=Label(frame,text="Date Of Birth")
dobLabel.grid(row=9,column=0)
dobEntry=Entry(frame,textvariable=dob)
dobEntry.grid(row=9,column=1,pady=5)
genderLabel=Label(frame,text='Gender')
genderLabel.grid(row=10,column=0)
gender=Combobox(frame,value=['Male','Female'],font='arial',width=25)
gender.grid(row=10,column=1,pady=5)
perAdd=Label(frame,text="Permanent Address & Contact Details").grid(row=12,column=1,pady=30)
locLabel=Label(frame,text='locality')
```

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```
locLabel.grid(row=13,column=0)
locEntry=Entry(frame,textvariable=loc,width=60)
locEntry.grid(row=13,column=1,pady=5)
pinLabel=Label(frame,text='P.I.N.')
pinLabel.grid(row=14,column=0)
pinEntry=Entry(frame,textvariable=pin,width=30)
pinEntry.grid(row=14,column=1,pady=5)
distLabel=Label(frame,text='District')
distLabel.grid(row=15,column=0)
distEntry=Entry(frame,textvariable=dist,width=30)
distEntry.grid(row=15,column=1,pady=5)
stateLabel=Label(frame,text='State')
stateLabel.grid(row=16,column=0)
stateEntry=Entry(frame,textvariable=state,width=30)
stateEntry.grid(row=16,column=1,pady=5)
countryLabel=Label(frame,text='Country')
countryLabel.grid(row=17,column=0)
countryEntry=Entry(frame,textvariable=country,width=30)
countryEntry.grid(row=17,column=1,pady=5)
emailLabel=Label(frame,text='E-Mail Address')
emailLabel.grid(row=18,column=0)
emailEntry=Entry(frame,textvariable=emAd,width=60)
emailEntry.grid(row=18,column=1,pady=5)
phoneLabel=Label(frame,text='Phone Number')
phoneLabel.grid(row=19,column=0)
phoneEntry=Entry(frame,textvariable=phn,width=60)
phoneEntry.grid(row=19,column=1,pady=5)
dietLabel=Label(frame,text='Dietary Preferance')
dietLabel.grid(row=11,column=0,pady=20)
diet=Combobox(frame,value=['Veg','Non-Veg'],font='arial',width=25)
```

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```
diet.grid(row=11,column=1,pady=5)
parentname=Label(frame,text="Parents' Details").grid(row=20,column=1,pady=30)
fatherLabel=Label(frame,text="Father's Name")
fatherLabel.grid(row=21,column=0)
fatherEntry=Entry(frame,textvariable=father,width=60)
fatherEntry.grid(row=21,column=1,pady=5)
motherLabel=Label(frame,text="Mother's Name")
motherLabel.grid(row=22,column=0)
motherEntry=Entry(frame,textvariable=mother,width=60)
motherEntry.grid(row=22,column=1,pady=5)
gaurdLabel=Label(frame,text="Gaurdian Name")
gaurdLabel.grid(row=23,column=0)
gaurdEntry=Entry(frame,textvariable=gaurd,width=60)
gaurdEntry.grid(row=23,column=1,pady=5)
parentadd=Label(frame,text="Parents' Address & Contact Details").grid(row=24,column=1,pady=30)
parentslocLabel=Label(frame,text='locality')
parentslocLabel.grid(row=24,column=0)
parentslocEntry=Entry(frame,textvariable=p_loc,width=60)
parentslocEntry.grid(row=24,column=1,pady=5)
parentspinLabel=Label(frame,text='P.I.N.')
parentspinLabel.grid(row=25,column=0)
parentspinEntry=Entry(frame,textvariable=p_pin,width=30)
parentspinEntry.grid(row=25,column=1,pady=5)
parentsdistLabel=Label(frame,text='District')
parentsdistLabel.grid(row=26,column=0)
parentsdistEntry=Entry(frame,textvariable=p_dist,width=30)
parentsdistEntry.grid(row=26,column=1,pady=5)
parentsstateLabel=Label(frame,text='State')
parentsstateLabel.grid(row=27,column=0)
parentsstateEntry=Entry(frame,textvariable=p_state,width=30)
```

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```
parentsstateEntry.grid(row=27,column=1,pady=5)
parentscountryLabel=Label(frame,text='Country')
parentscountryLabel.grid(row=28,column=0)
parentscountryEntry=Entry(frame,textvariable=p_country,width=30)
parentscountryEntry.grid(row=28,column=1,pady=5)
parentsemailLabel=Label(frame,text='E-Mail Address')
parentsemailLabel.grid(row=29,column=0)
parentsemailEntry=Entry(frame,textvariable=p_emAd,width=60)
parentsemailEntry.grid(row=29,column=1,pady=5)
parentsphoneLabel=Label(frame,text='Phone Number')
parentsphoneLabel.grid(row=30,column=0)
parentsphoneEntry=Entry(frame,textvariable=p_phn,width=60)
parentsphoneEntry.grid(row=30,column=1,pady=5)
foccuLabel=Label(frame,text="Father's Occupation")
foccuLabel.grid(row=31,column=0)
foccuEntry=Entry(frame,textvariable=foccu,width=60)
foccuEntry.grid(row=31,column=1,pady=5)
moccuLabel=Label(frame,text="Mother's Occupation")
moccuLabel.grid(row=32,column=0)
moccuEntry=Entry(frame,textvariable=moccu,width=60)
moccuEntry.grid(row=32,column=1,pady=5)
EduLabel=Label(frame,text="Educational Details(CLASS 12th)").grid(row=33,column=1,pady=30)
e_marks=Label(frame,text='Marks in English')
e_marks.grid(row=34,column=0)
e_entry=Entry(frame,textvariable=e)
e_entry.grid(row=34,column=1)
p_marks=Label(frame,text='Marks in Physics')
p_marks.grid(row=35,column=0)
p_entry=Entry(frame,textvariable=p)
p_entry.grid(row=35,column=1)
```


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```
c_marks=Label(frame,text='Marks in Chemistry')
c_marks.grid(row=36,column=0)
c_entry=Entry(frame,textvariable=c)
c_entry.grid(row=36,column=1)
m_marks=Label(frame,text='Marks in Maths')
m_marks.grid(row=37,column=0)
m_entry=Entry(frame,textvariable=mth)
m_entry.grid(row=37,column=1)
op_marks=Label(frame,text='Marks in Optional')
op_marks.grid(row=38,column=0)
op_entry=Entry(frame,textvariable=op)
op_entry.grid(row=38,column=1)
othDet=Label(frame,text="Other Details").grid(row=39,column=1,pady=30)
hobb=Label(frame,text='Hobbies(Optional)')
hobb.grid(row=40,column=0)
hobb=Entry(frame,textvariable=hobbies)
hobb.grid(row=40,column=1)
ach=Label(frame,text='Any Achievement(Optional)')
ach.grid(row=41,column=0)
ach=Entry(frame,textvariable=achieve)
ach.grid(row=41,column=1)
Button(frame,text='Submit',command=submit).grid(row=60,column=2,pady=50)
Button(frame,text='Clear',command=clear).grid(row=60,column=3,padx=25,pady=50)
Button(frame,text='Exit',command=lambda:window.destroy()).grid(row=60,column=4,padx=15,pady=50)
window.mainloop()
```

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Admission Form

Fill out the Admission Form

Student's Details

First Name

Middle Name

Last Name

Date Of Birth

Gender

Dietary Preference

Permanent Address & Contact Details

locality

P.I.N.

District

State

Country

E-Mail Address

Phone Number

Parents' Details

Father's Name

Mother's Name

Gaurdian Name

Admission Form

Permanent Address & Contact Details

locality

P.I.N.

District

State

Country

E-Mail Address

Phone Number

Parents' Details

Father's Name

Mother's Name

Gaurdian Name

locality

P.I.N.

District

State

Country

E-Mail Address

Phone Number

Father's Occupation

Mother's Occupation

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Admission Form

Guardian Name: GGGGG

Locality: LLLLLL

P.I.N: 12121212

District: D1D1D1D1D1

State: STATE

Country: CO222

E-Mail Address: EMAIL.ADDD@EMAIL.COM

Phone Number: 123432

Father's Occupation: SBDGE

Mother's Occupation: ERYDBS

Educational Details(CLASS 12th)

Marks in English: 098

Marks in Physics: 075

Marks in Chemistry: 075

Marks in Maths: 050

Marks in Optional: 0100

Other Details

Hobbies(Optional): WERTY

Any Achievement(Optional): ASDFG

Submit Clear Exit

AutoSave OFF MicroProject1.data • Saved to this PC

MANAV KUMAR

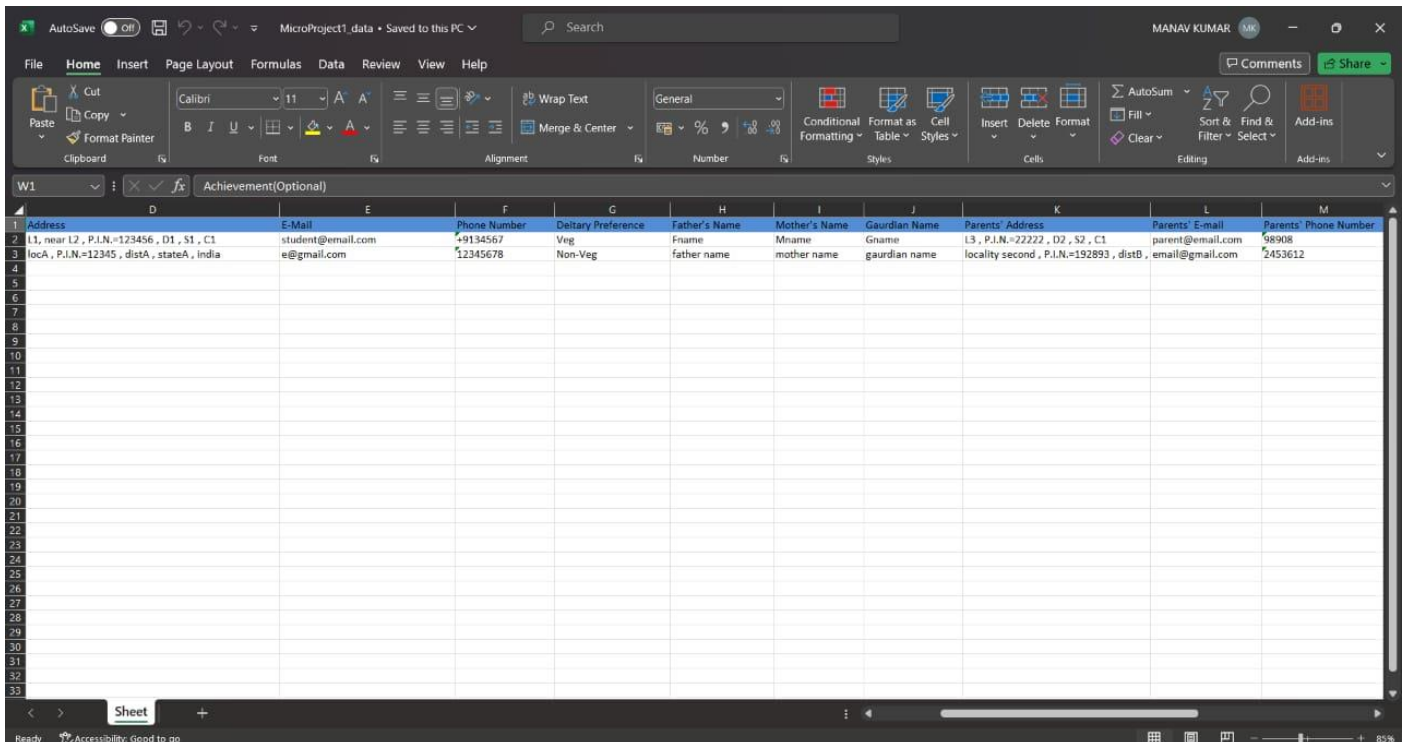
File Home Insert Page Layout Formulas Data Review View Help

Clipboard Font Alignment Number Styles Cells Editing Add-ins

	L	M	N	O	P	Q	R	S	T	U	V	W
	Parents' E-mail	Parents' Phone Number	Father's Occupation	Mother's Occupation	English Marks	Physics Marks	Chemistry Marks	Maths Marks	Optional 1 Marks	Aggregate Marks(PCM)	Hobbies(Optional)	Achievement(Optional)
1	parent@email.com	98908	Business	Homemaker	98	75	75	50	99	66.66666667	Reading	Nil
2	email@gmail.com	2453612	foccu	moccu	75	57	78	98	76	77.66666667	reading	nil
3												
4												
5												
6												
7												
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9												
10												
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33												

Sheet

Ready Accessibility: Good to go



4. DISCUSSION AND CONCLUSION

In conclusion, we have successfully developed a GUI (Graphical User Interface) for a student admission form. This form simplifies the process of collecting all the necessary information from students and eliminates the need for manual paperwork. The GUI provides a user-friendly interface and allows students to easily fill out their personal and academic details. By creating this GUI, we have made the admission process more efficient and time-saving. The form ensures that all required information is captured accurately and securely. It also eliminates the chances of errors that can occur during manual data entry. Additionally, the GUI can be customized and extended to accommodate specific requirements of different educational institutions. The fields and sections in the form can be modified based on the admission criteria and preferences of each institution.

5.FUTURE SCOPE

Although we have successfully created a student admission form GUI, there are several avenues for future improvements and enhancements. Here are some areas that can be explored:

- Database Integration: Currently, the form captures information but does not store it in a database. In the future, integrating a database system can allow for efficient management and retrieval of student records.
- Validation and Error Handling: Enhancing the GUI to include validations and error handling mechanisms can help ensure that students provide valid and accurate information while filling out the form.
- Integration with Online Payment: Adding a payment gateway integration will enable students to pay their admission fees securely and conveniently.
- Data Analytics and Reporting: Implementing data analytics features can provide valuable insights into student admission trends and help in making informed decisions. It can also help generate customized reports for administrative purposes.
- User Authentication and Security: Enhancing the GUI with user authentication mechanisms and implementing secure data storage can protect sensitive student information.

By continuously improving and expanding its features, we can make the process even more efficient and user-friendly for both students and educational institutions

ANNEXURE: REFERENCE AND BIBLIOGRAPHY

- ❖ *Python GUI Programming with Tkinter* by ALAN D. MOORE (Second Edition)
- ❖ *Python Programming* by REEMA THAREJA (Third Edition)

